### IV.2.1.3. Excision Margins

The width of the healthy tissue margin removed around a cutaneous squamous cell carcinoma (SCC) is a critical factor in achieving complete tumour removal and remains a central issue in surgical management.

According to the 2002 guidelines from the British Association of Dermatologists (BAD) and the National Comprehensive Cancer Network (NCCN), data from Brodland and Zitelli[125]—based on a prospective study of 141 primary SCCs treated with Mohs micrographic surgery—support the following:

- A **4-mm margin** is adequate for complete eradication in **95% of tumours <2 cm** in diameter.
- For **tumours ≥2 cm** or those with high-risk features (e.g., poor differentiation, subcutaneous invasion, or location in high-risk areas), a margin of **>6 mm** is required to achieve similar clearance rates (level of evidence 2).

In certain higher-risk scenarios, margins should be increased to **10 mm or more** to ensure comparable local control. These include:

- Recurrent tumours requiring re-excision after positive margins on prior surgery
- High histological grade (poorly differentiated SCC)
- Deep invasion into the hypodermis
- Tumour location in anatomically high-risk areas (e.g., face, ears, lips)
- Presence of **perineural invasion**
- Large tumour size with potential for subclinical spread

For very large lesions, even wider margins may be considered.

#### *Recommendations*

Given that tumour size alone is an imprecise indicator of biological aggressiveness and that standard margins fail in approximately 5% of cases, the working group recommends tailoring excision margins according to the prognostic group of the SCC:

- **Group 1 tumours (low risk):** A standardized excision margin of **4–6 mm**, with standard histopathological processing and macroscopic sampling to maximize margin assessment.
- **Group 2 tumours (high risk):** Extended margins of **≥6 mm**, and often **10 mm or more**, particularly when multiple risk factors for subclinical extension are present (professional consensus).

### *Excision Depth*

There is less consensus regarding optimal depth of excision. However, the following principles apply:

- Full-thickness excision should include the **hypodermis** to ensure complete tumour removal.
- Deeper structures such as **aponeuroses, periosteum, and perichondrium** should be preserved unless there is clinical or histological evidence of tumour involvement.

### *Margin Assessment Techniques*

The reliability of margin evaluation methods varies:

- **Extemporaneous (frozen section) examination** can be valuable when performed by the surgeon, particularly when focused on areas at highest risk for residual disease. It should be as thorough as possible in these regions to improve detection of involved margins.